Hiteshkumar Gupta

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SUMMARY

Experienced Full Stack Developer with 3 years in Python and JavaScript, specializing in Django and React. Proven expertise in scalable architecture design, RESTful API development, PostgreSQL schema optimization, and AWS deployment. Skilled in optimizing React frontends with Redux to enhance user experience and developing AI-powered systems using Retrieval-Augmented Generation, data pipelines, and large language models for advanced data retrieval. A strong advocate for high-quality, well-documented, and tested code, collaborating effectively to deliver exceptional software solutions.

TECHNICAL SKILLS

Programming Languages: Python, Java, JavaScript, TypeScript, Kotlin, C++, HTML, CSS
Web and Mobile Stack: NodeJS, ReactJS, Django, Angular, Spring Boot, Android, Firebase
Database: MySQL, Postgres, MongoDB, DynamoDB, Firebase Realtime Database
Cloud Tools: AWS (DynamoDB, S3, Lambda, API Gateway, CloudFormation, and CloudFront), GCP (Kubernetes, Terraform)
Dev and ML Tools: Git, VS Code, TDD, CI/CD, Linux, NPM, NVM, Yarn, Jenkins, Jest, SEO, TensorFlow, PyTorch, Docker
Soft Skills: Problem-Solving, Decision making, Leadership, Communication, Collaboration

WORK EXPERIENCE

SIEMENS ***Saskatoon, Canada***
SOFTWARE ENGINEER - AI Sept. 2023 – Current

- Contributed to an innovative **Retrieval-Augmented Generation** system integrating advanced data pipelines, vector embeddings, and **large language models**, enabling data retrieval.
- Developed and optimized **React** components to improve the user interface and experience of AI-powered applications, ensuring smooth and responsive interactions.
- Built and integrated **FastAPI** endpoints to facilitate efficient communication between front-end systems and AI models, optimizing performance and reducing latency.
- Assisted in streamlining **DevOps** processes using Jenkins, automating build and deployment pipelines for continuous integration and delivery, improving system reliability and efficiency.
- Conducted **performance optimizations** on AI systems and APIs, improving response times and scalability to handle large-scale data processing.
- Participated in **code reviews** and **team discussions** to maintain high standards of coding practices, promoting teamwork and knowledge sharing within the development team.
- Skills: RAG, React, FastAPI, Jenkins, LLM Integration, DevOps, AI Systems, Collaboration

Manentia AI ***Delhi, India***
SOFTWARE ENGINEER Aug. 2022 – Jul. 2023

- Developed and implemented a new **microservices architecture** using **Python** and **Docker** for deployment, resulting in a 40% increase in system reliability and a 25% reduction in bug-related delays.
- Built RESTful **Django** APIs using Test-Driven Development methodologies, ensuring robustness and reliability of codebase.
- Created and optimized complex queries using **Django ORM** to efficiently retrieve, manipulate, and aggregate data, ensuring minimal database load and enhancing overall application performance.
- Developed custom **middleware** and **signal handlers** to extend Django's functionality and improve application performance.
- Optimized **PostgreSQL** database schemas and wrote efficient queries, reducing query execution time by 40%.
- Optimized **React frontend**, significantly reducing load time by **25%** through **code-splitting** and **lazy loading techniques**, enhancing user experience.
- Migrated to **AWS cloud** solution, deploying infrastructure using CloudFormation, reducing deployment time by 10%
- Utilized **AWS Lambda** for serverless computing, reducing infrastructure costs by 20%.
- Monitored applications using **AWS CloudWatch**, enhancing system reliability and performance.
- Environment: Django, Postgres, React, Redux, Docker, AWS Infrastructure, Leadership, Communication, Collaboration

Mobiuso Technologies Pvt. Ltd. ***Ahmedabad, India***
SOFTWARE ENGINEER Apr. 2021 – Aug. 2022

- Developed and tested RESTful **Django APIs** with consistent coding patterns for high code reliability and performance.
- Enhanced **NodeJS API** responsiveness with asynchronous views and database caching, resulting in substantial performance improvements.
- Developed custom middleware and event handlers to extend **Express.js** functionality and improve application performance.
- Reduced React project load time by **30%** with webpack optimization, and bundle analysis, enhancing website performance.
- Implemented advanced web scrapers on **AWS Linux Servers**, enhancing data accuracy and enabling data-driven decision-making for clients through extraction, processing, and detailed storage in Excel for easy filtering.
- Developed advanced **React** components for the clients, greatly enhancing their user interfaces and ensuring exceptional quality, with a 95% client satisfaction rate.
- Skills: Django, NodeJS, ExpressJS, React, Linux, Data Processing and Analysis, Communication, Innovative, Teamwork

EDUCATION

Dalhousie University ***Halifax, Canada***
Master of Applied Computer Science (GPA – 4.2 / 4.3) Sep. 2023 – Dec. 2024

Parul University ***Vadodara, India***
Bachelor of Technology in Computer Science and Engineering (GPA – 4.0 / 4.3) Jun. 2017 - May 2021

PROJECTS

Cleanliness Automation: YoloV3 (*YoloV3, Tensorflow, Keras, Flask*) GitHub Link
- IEEE Recognized, Waste Detection App – an innovative solution for identifying and categorizing real-time waste using ML.
- Utilized TensorFlow framework and YoloV3 base architecture to build 256 CNN-layer model.
- Integrated model With a clean and intuitive Flask-based web interface for interaction.

QuickHire (*ReactJS, NodeJS, MongoDB, Stripe, CI/CD, Git*) Website / GitHub Link
- QuickHire, an online marketplace where people can offer their skills and services for others to hire or buy.
- Built UI and RESTful APIs for seamless user authentication and profile management using OAuth2.0 and JWT token.
- Crafted user-focused features like Wishlist and feedback systems, paired with efficient service listing and search functionalities, enhancing overall user experience and exploration, using UML for system design, boosting trust and satisfaction.

RentX (*ReactJS, SpringBoot, MySQL, Stripe, CI/CD, Git*) GitHub Link
- RentX, an online marketplace for buying and selling items built with Java SpringBoot, and ReactJS and used MySQL.
- Crafted seamless user interactions including registration, profiles, search, and Wishlists, enhancing user engagement.
- Used agile methodology to improve Rent X with responsive design and feature upgrades based on user and market insights.

DalCommunity Mobile App (*Kotlin, Firebase*) GitHub Link
- Dal Community App facilitates seamless interaction and information exchange among students and faculties.
- Enhanced user engagement by implementing features like scheduling and announcements
- Enabled inclusivity and activism among users by integrating features such as anonymous posting, petitions, polls, event coordination, and cause support, while ensuring real-time data storage and retrieval using the Firebase Realtime Database.

SparkMedia Insights (*Java, GCP Dataproc, Hadoop, Map, Reduce, Spark*) GitHub Link
- Preprocessed large news data using GCP Dataproc, removing HTML tags, special symbols, and stop words.
- Developed a Spark application to tokenize text and map words to counts using the Map function.
- Aggregated word counts with the Reduce function and analyzed the results.

PUBLICATION

Title: Cleanliness Automation: YoloV3 *DOI: 10.1109/I2CT51068.2021.9418056* Paper Link
Publisher: IEEE at *2021 6th International Conference for Convergence in Technology* Apr. 2021
Role: Author, Researcher, Executor
- Conducted comprehensive research and dataset preparation for real-time waste detection.
- Implemented YOLOv3 after rigorous experimentation, achieving 98% accuracy.
- Integrated YOLOv3 into Flask with FastAPI, demonstrating real-world applicability and driving advancements in monitoring and automation.

CERTIFICATIONS

Getting Started with AWS Machine Learning Certificate Link
- Acquired familiarity with AWS ML services including Rekognition, Textract, and SageMaker along with EC2, and VPC.

Angular - The Complete Guide (2024 Edition) Certificate Link
- I have learned to develop modern, complex, and scalable web applications with Angular, mastering components, HTTP requests, routing, authentication, forms, TypeScript, and application architecture.

Google Developer Student Clubs Certificate Link
- Earned Google Cloud badges in network setup, Kubernetes, ML integration, and data engineering.

Introduction to Machine Learning from NPTEL Certificate Link
- Gained knowledge in essential machine learning algorithms, computational learning theory, feature reduction methods, and hands-on Python programming, including practical problem-solving

ACHIEVEMENTS

- Enjoy solving coding problems focusing on data structures, algorithms, and database concepts (Leetcode).
- Achieved 3rd place in a GitHub competition held at Parul University.
- Contributed articles on Machine Learning and Artificial Intelligence to dockship.io and medium.com.
- Awarded a medal of excellence for exceptional performance throughout Bachelor's degree studies.
- Participated and secured 4th rank in the Gen AI 2024 Hackathon, contributing to a team project focused on developing an innovative AI-driven healthcare product (DevPost Hackathon).